School of Environment and Society
Department of Architecture and Building Engineering

Master's Thesis

Structural Harmony in Equestria: Load-Bearing Friendship and the Architecture of Ponyville

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2026

Abstract

This thesis investigates *(state the problem and your approach in one or two sentences)*. *(Summarise the method.) (Summarise the key results, with the single most important number.) (State the main conclusion and why it matters.)*

Keep the abstract to a single page: motivation, method, results, and contribution, with no citations and no undefined acronyms.

Keywords: keyword one; keyword two; keyword three; keyword four

Acknowledgment

To be written in the final draft.

I would like to express my sincere gratitude to my supervisor, Prof. Masashi Matsuoka, for *(. . .)*. I also thank the members of the Matsuoka Laboratory for *(. . .)*, and my family for their support.

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List of Abbreviations

ISCT Institute of Science Tokyo

GIS Geographic Information System

RMSE Root Mean Square Error

1 Introduction

1.1 Background and Motivation

(Set the scene. What is the real-world problem, and why does it matter? Give the reader the context they need in a few paragraphs, and cite the key prior work [1]. End the section by narrowing from the broad context down to the specific gap your thesis addresses.)

1.2 Problem Statement

(State precisely the problem you tackle. What is unknown, missing, or unsatisfactory in current approaches? Make the gap concrete and specific.)

1.3 Research Questions and Objectives

(State the objective of the thesis in one sentence, then list the concrete questions it answers.)

1. *(First research question.)*
2. *(Second research question.)*
3. *(Third research question.)*

1.4 Contributions

(List what is new in your work. Each bullet should be a concrete, defensible claim that the rest of the thesis supports.)

- *(First contribution.)*
- *(Second contribution.)*
- *(Third contribution.)*

1.5 Thesis Outline

(One short paragraph mapping the rest of the document.) Chapter 2 reviews the related work; Chapter 3 describes the data and materials; Chapter 4 presents the method; Chapter 5 reports and discusses the results; and Chapter 6 concludes and outlines future work.

2 Background and Related Work

(Review the prior work your thesis builds on, grouped into a few themed sections. The goal is to lead the reader naturally to the gap you identified in section 1.2.)

2.1 First Theme

(Summarise one strand of related work. Cite as you go — a single source [1] or several at once [1], [2]. Group papers by idea, not one paragraph per paper.)

2.2 Second Theme

(A second strand. Compare and contrast approaches: say what they do well, and where they fall short for your problem.)

2.3 Research Gap

(Synthesise the review into the specific gap this thesis fills. This section motivates the method in chapter 4.)

3 Data and Materials

(Describe the data, datasets, study areas, or materials you use. State where they come from, how large they are, and how they are split or prepared. The table below is a worked example you can copy and adapt.)

3.1 Dataset Overview

(Introduce each dataset and its role.) A summary is given in table 3.1.

Table 3.1. Example dataset summary. Replace with your own data.

Property	Dataset A	Dataset B
Role	training + test	held-out test
Samples	10,000	100
Source	*(source)*	*(source)*

3.2 Preprocessing

(Explain how the raw data is cleaned, aligned, normalised, or otherwise transformed before it is used in chapter 4.)

4 Methodology

(Present your method in enough detail that a competent reader could reproduce it. Start with an overview, then go component by component. The figure and equation below are worked examples.)

4.1 Overview

(One or two paragraphs and a diagram giving the big picture before the details.) Figure 4.1 shows the overall pipeline.

Figure 4.1. The proposed method, as modelled by its mascot. Swap this for your real overview diagram before submitting (keeping a pony in your drafts for morale is, however, fully endorsed).

4.2 Proposed Approach

(Describe the core of your method. Define your notation, and number any important equation so you can refer back to it, as in eq. (4.1)*.)*

$$y = f(x; \theta). \tag{4.1}$$

4.3 Experimental Setup

(Training details, hyper-parameters, baselines, and the evaluation metrics and protocol — everything the reader needs to interpret chapter 5.)

We report results with the Root Mean Square Error (RMSE) over a Geographic Information System (GIS) study area; full numbers are in chapter 5. (The first use of an acronym defined in `frontmatter/abbreviations.tex` expands automatically, like Institute of Science Tokyo (ISCT).)

5 Results and Discussion

(Report your findings in a logical order, then interpret them. Lead with the headline result, support each claim with a table or figure, and discuss what it means — not just what the number is.)

5.1 Main Results

(Present the headline comparison.) Table 5.1 reports an example.

Table 5.1. Example results table. Replace with your own numbers.

Method	Metric 1 ↑	Metric 2 ↓
Baseline	0.71	0.34
Proposed	**0.86**	**0.21**

5.2 Discussion

(Interpret the results. Why do they come out this way? What are the limitations and threats to validity? Connect the findings back to the research questions in section 1.3.)

6 Conclusion and Future Work

6.1 Summary of Findings

(Restate, briefly, what you set out to do and what you found. Answer the research questions from section 1.3 directly.)

6.2 Contributions Revisited

(Tie the findings back to the contributions you claimed in section 1.4.)

6.3 Limitations

(State honestly what bounds your conclusions — the assumptions, scope, and caveats a careful reader should keep in mind.)

6.4 Future Work

(Where should the next student pick up? Give concrete, promising directions rather than vague aspirations.)

References

[1] J. Smith and J. Doe, "An example journal article for the thesis template," *Journal of Example Studies*, vol. 12, no. 3, pp. 100–120, 2024. DOI: `10.0000/example.2024.0123`

[2] T. Tanaka and H. Yamada, "An example conference paper," in *Proc. International Conference on Examples (ICE)*, 2023, pp. 1–8.

[3] M. Lee et al., *An example preprint*, arXiv preprint arXiv:2501.00000, 2025.

A Supplementary Material

(Optional. Put material here that supports the thesis but would interrupt the main text: extra tables and figures, full derivations, additional results, questionnaires, or implementation details.)

A.1 Code listing example

Use the `lstlisting` environment for code or algorithms, and reference it with listing A.1. The style (monospace, line numbers, thin rules) is set once in the class file.

```
def harmony(friends):
    """Return the friendship coefficient of a group of ponies."""
    return sum(f.magic for f in friends) / len(friends)
```

Listing A.1. A minimal example listing.

To pull in a whole source file instead, use the `\lstinputlisting` command.

B LaTeX Feature Reference

This appendix is a template cheat-sheet — delete it before submission.

B.1 Citations

Cite one source [1] or several [1], [2], [3]. Entries live in `references.bib`; the style is IEEE numeric.

B.2 Figures

Reference floats by label: fig. B.1.

Figure B.1. Caption goes *below* a figure. Put image files in `figures/` and always add a `\label`.

B.3 Subfigures (side-by-side)

Place two images in one figure, each with its own sub-caption. Reference the whole figure with fig. B.2, or a single panel with fig. B.2b.

(a) One pony. **(b)** Another pony.

Figure B.2. Two images side by side, each with its own sub-caption.

B.4 Tables

Use `booktabs` rules (no vertical lines), caption *above*, as in table B.1.

Table B.1. Example results table.

Method	Metric 1	Metric 2
Baseline	0.71	0.34
Proposed	**0.86**	**0.21**

B.5 Equations

Number important equations and reference them, e.g. eq. (B.1):

$$y = f(x; \theta). \tag{B.1}$$